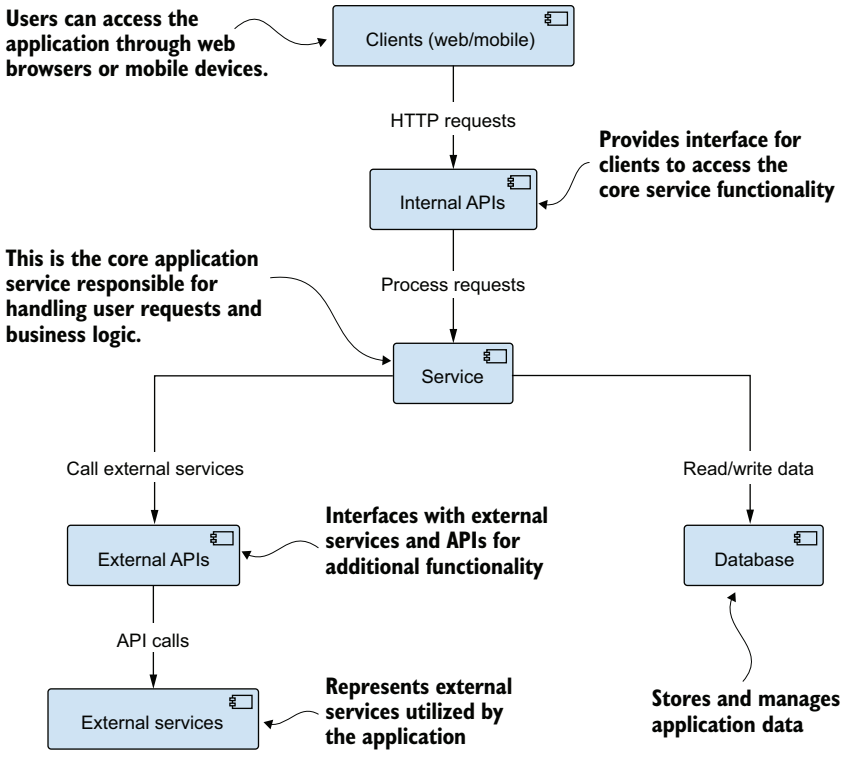


Figure 1.3 Simplified architecture diagram of a web application ecosystem. Clients, such as web browsers and mobile devices, interact with the core application service, which handles user requests and implements business logic. The service interacts with a database to store and manage application data. Additionally, the service communicates with external APIs to access additional functionality and interacts with external services utilized by the application.

licensed and open source versions. Figure 1.4 illustrates the interconnected components of a generative AI web application and the tools we'll use to build the examples in this book.

As we build our apps, we will use JavaScript and React to display our UI interface components, along with Next.js and the Vercel AI SDK to manage the backend and interact with external AI service providers. While the Vercel AI SDK supports various providers, our projects will primarily use Google Gemini models by default, with occasional use of OpenAI models.

React is one of the most popular JavaScript libraries for building user UIs with reusable components. It's easy to learn and has vast support for AI-related components. We will use React to build our UI elements and to present them to the users with accessibility and performance in mind.

Next.js is a framework built on React that simplifies the development of server-rendered and statically generated web applications. It streamlines backend management